



POLYDISPERSITY INDEX AND MOLECULAR WEIGHT DISTRIBUTIONS OF POLYMERS

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(Received 3 July 1995; accepted in final form 14 November 1995)

Abstract—The polydispersity index as a measure of the width of molecular weight distributions (MWD) is theoretically very important. However, considering MWDs of real polymers, its importance is partly lost. Consequently, several problems arise. The most common one is the misinterpretation of polydispersity index as a single and unique measure of the MWD width. In this work, the origin and solutions of the problem are clearly described via several theoretical distribution functions, i.e. those which are most widely applied for the description of polymer MWDs. The conclusions are confirmed by the analysis of MWDs of real samples, namely styrene–acrylonitrile copolymers. In addition, a complete procedure for the MWD data analysis is presented. Such a procedure is necessary for the correct interpretation and comparison of different, experimentally obtained, molecular weight distributions of polymers. Copyright © 1996 Elsevier Science Ltd

NOMENCLATURE

- A Area under the distribution curve.
- $B()$ Experimentally determined relative frequencies of molecules in the polymeric sample.
- b Parameter of the Schulz distribution function.
- $F()$ Differential form of frequency distribution.
- $I_v()$ Integral form of frequency distribution.
- $I_w()$ Integral form of mass distribution.
- M Molecular weight.
- M_0 Molecular weight of repeating unit in polymer.
- $\bar{M}_n$ Number average molecular weight.
- $\bar{M}_w$ Weight average molecular weight.
- m Parameter of the Tung distribution function.
- p Parameter of the Flory distribution function.
- p Parameter of the Schulz distribution function.
- x Degree of polymerization.
- $\bar{x}_n$ Number average degree of polymerization.
- $\bar{x}_m$ Parameter of the logarithmic-normal distribution function.
- $\bar{x}_w$ Weight average degree of polymerization.
- y Parameter of the Tung distribution function.
- $W()$ Differential form of mass distribution.
- α Parameter of the Schulz distribution function.
- Γ Complete gamma-function.
- γ Incomplete gamma-function.
- σ Parameter of the logarithmic-normal distribution function.
- σ_s^2 Variance of the molecular weight distribution.

INTRODUCTION

As it is well known, the incorporation of monomer units into polymer molecules during polymerization reaction follows certain statistical principles. Consequently, the product of such reaction is never homogeneous at the molecular level, i.e. molecular properties are not constant throughout the polymer sample. Therefore, while discussing a molecular property of polymer, one has to keep in mind the

existence of related distribution, which links discrete property values with corresponding frequencies of molecules bearing those property values. Depending on the polymer type [1], a sample may be polydisperse in reference to microstructure, degree of chain branching, composition of copolymer, sequence of comonomer units in copolymer, etc. Besides, almost every polymer is polydisperse regarding the molecular weight. One must remember that specific properties of the polymeric material and the polymeric character itself are determined, among other things, by the unusually large molecular size. Accordingly, the changes in the shape and width of molecular weight distributions are directly reflected in the changes of various properties. Therefore, the basis of polymer characterization has to be the determination of molecular weight distribution (MWD). Such determinations are nowadays performed routinely, by means of well-known size exclusion chromatography (SEC) method. Many excellent review articles [2–4] describing the general features of SEC may be found in the literature. Great importance is given to the interpretation of experimental data, usually with the aid of modern computer simulation techniques [5–10]. Still, the way of data presentation and the quality of result interpretation often fall behind the possibilities of modern experimental techniques, which results in loss of information. In this short review, attention will be directed towards the unfortunately very common misinterpretation of the polydispersity index, which is used for the description of the MWD. Also, a point will be given to the procedures that enable a quantitative comparison of different molecular weight distributions, and which should be performed in every MWD data analysis.

A large part of the following discussion is not necessarily restricted to the investigation of polymer

MWD, and some presented solutions can be successfully transferred to other fields of chemical engineering, e.g. wherever a problem of particle size distribution is encountered.

BASIC DEFINITIONS

A polymeric molecule can often be viewed as a long chain of interconnected repeating units. The number of such units, x , is called the degree of polymerization. The frequency distribution defines the fraction of molecules with certain degree of polymerization. The sum of all frequencies equals 1:

$$\sum_{x=1}^{\infty} F(x) = 1. \quad (1)$$

The degree of polymerization is usually much larger than 1, and it can take any of the whole-numbered values between the minimal and maximal. As a consequence, a discrete distribution can often be approximated by the continuous one, $F(x)dx$. The sum of frequencies is replaced by the integral:

$$\int_0^{\infty} F(x)dx = 1. \quad (2)$$

Except in the differential form, the degree of polymerization distributions (DPD) can be presented in the integral form, as the sum:

$$I_x(x) = \sum_{x=1}^x F(x) \quad (3)$$

or as the integral:

$$I_x(x) = \int_0^x F(x)dx \quad (4)$$

where $I_x(x)$ denotes the cumulative fraction of molecules with the degree of polymerization less than or equal to x .

Obviously, the degree of polymerization distributions belong to the family of probability functions, and the starting point for their description is the mean of the function. The mean of the DPD is called the number average degree of polymerization. The cumulative fraction of molecules with the degree of polymerization less than the number average is exactly 0.5. The DPD number average is determined by the following relation:

$$\bar{x}_n = \frac{\sum_{x=1}^{\infty} xF(x)}{\sum_{x=1}^{\infty} F(x)} \quad (5)$$

or by

$$\bar{x}_n = \frac{\int_0^{\infty} xF(x)dx}{\int_0^{\infty} F(x)dx} \quad (6)$$

The relative contribution of molecules to some polymer property can often be proportional not to their number, but rather to their mass fractions, so the larger molecules gain the greater significance. Therefore a different measure of the DPD average should be postulated—the weight average degree of polymerization. The cumulative mass fraction of molecules with the degree of polymerization less than the weight average is 0.5. The DPD weight average is defined by the following expression:

$$\bar{x}_w = \frac{\sum_{x=1}^{\infty} x^2 F(x)}{\sum_{x=1}^{\infty} xF(x)} \quad (7)$$

or

$$\bar{x}_w = \frac{\int_0^{\infty} x^2 F(x)dx}{\int_0^{\infty} xF(x)dx} \quad (8)$$

In the experimental determination of the MWD of polymers, the detector response can often be directly proportional to the mass fraction of molecules, and not to the number fraction, which creates a need for defining mass distributions of polymerization degrees:

$$\sum_{x=1}^{\infty} W(x) = 1 \quad (9)$$

or

$$\int_0^{\infty} W(x)dx = 1 \quad (10)$$

related to the frequency distributions by

$$W(x) = \frac{xF(x)}{\sum_{x=1}^{\infty} xF(x)} \quad (11)$$

and

$$W(x)dx = \frac{x F(x) dx}{\int_0^{\infty} x F(x) dx} \quad (12)$$

The mass distributions can be given in the integral form:

$$I_w(x) = \sum_{x=1}^x W(x) \quad (13)$$

$$I_w(x) = \int_0^x W(x)dx \quad (14)$$

where $I_w(x)$ denotes the cumulative mass fraction of molecules with the degree of polymerization less than or equal to x .

Averages of the polymerization degrees are

calculated from mass distributions by application of the following equations:

$$\bar{x}_n = \frac{\sum_{x=1}^{\infty} W(x)}{\sum_{x=1}^{\infty} \frac{W(x)}{x}} \quad (15)$$

$$\bar{x}_n = \frac{\int_0^{\infty} W(x) dx}{\int_0^{\infty} \frac{W(x) dx}{x}} \quad (16)$$

$$\bar{x}_w = \frac{\sum_{x=1}^{\infty} xW(x)}{\sum_{x=1}^{\infty} W(x)} \quad (17)$$

$$\bar{x}_w = \frac{\int_0^{\infty} xW(x) dx}{\int_0^{\infty} W(x) dx} \quad (18)$$

To calculate molecular weight averages, the degree of polymerization, appearing in equations (1)–(18), should be replaced by the product

$$M = x \cdot M_0 \quad (19)$$

where M_0 stands for molecular weight of the repeating unit in the polymer molecule. Thus, the most frequently applied equation for calculation of the number average molecular weight is obtained in the following form:

$$\bar{M}_n = \frac{\sum_{M=M_0}^{\infty} W(M)}{\sum_{M=M_0}^{\infty} \frac{W(M)}{M}} \quad (20)$$

and for the weight average molecular weight in the similar form:

$$\bar{M}_w = \frac{\sum_{M=M_0}^{\infty} MW(M)}{\sum_{M=M_0}^{\infty} W(M)} \quad (21)$$

The number and the weight average molecular weight are the most important central measures of the polymer MWD. Both averages may be calculated from the experimentally determined MWD, but may be measured directly as well. Beside these two basic mean values, one may define the higher averages, the most important of which is the so called z -average:

$$\bar{M}_z = \frac{\sum_{M=M_0}^{\infty} M^2 W(M)}{\sum_{M=M_0}^{\infty} MW(M)} \quad (22)$$

THEORETICAL DISTRIBUTION FUNCTIONS

In most cases the discrete MWD may be approximated by some suitable continuous theoretical distribution function. The benefit of those approximations is apparent. Theoretical distribution functions, which are completely defined by only few adjustable parameters, may be easily analyzed and later built into the variety of correlations. The commonly applied theoretical distribution functions may be divided into two main groups. The first contains the functions developed exclusively for the description of polymer MWD, using simple theoretical (statistical) considerations of the polymerization process (e.g. Flory, Schulz, Tung, Pearson, etc.). The other includes well-known probability functions capable of describing the real MWD (logarithmic-normal, Poisson function, etc.). All mentioned functions are unimodal, defined for positive arguments (degrees of polymerization), and asymmetric, i.e. inclined towards the lower x -values. As a consequence of the statistical approach generality, it may be proved that most of those functions present just special cases of the universal three-parameter exponential distribution function [11]. The following distributions are amongst the most frequently met theoretical functions.

(a) The one-parameter Flory distribution (parameter p), may be analytically presented in the differential form:

$$W(x) = x(\ln p)^2 p^x \quad (23)$$

or in the integral form:

$$I_w(x) = \frac{1}{p} - [1 + (1-p)x]p^{x-1} \quad (24)$$

and the corresponding number and weight averages are defined by the following equations:

$$\bar{x}_n = \frac{1}{1-p} \quad (25)$$

$$\bar{x}_w = \frac{1+p}{1-p}, \quad (26)$$

respectively.

(b) The two-parameter Schulz distribution, with parameters b and α (or p) is usually seen in one of the following forms:

$$W(x) = \frac{\alpha^{b+1} x^b}{\Gamma(b+1)} \exp(-\alpha x) \quad (27)$$

$$W(x) = \frac{(-\ln p)^{b+1} x^b p^x}{\Gamma(b+1)} \quad (28)$$

$$I_w(x) = \alpha^{b+1} e^{-\alpha x} \frac{\gamma(b+1, x)}{\Gamma(b+1)} \quad (29)$$

$$I_w(x) = \frac{(-\ln p)^{b+1}}{p} \frac{\gamma(b+1, x)}{\Gamma(b+1)} \quad (30)$$

where Γ and γ denote the complete and the

incomplete gamma-function, respectively. The relation between the parameters α and p is

$$\alpha = -\ln p \quad (31)$$

and the related number and weight averages are given by

$$\bar{x}_n = \frac{b}{\alpha} \quad (32)$$

$$\bar{x}_w = \frac{b+1}{\alpha}, \quad (33)$$

respectively.

(c) The two-parameter Tung distribution (parameters m and y) is shown in the subsequent analytical forms:

$$W(x) = myx^{m-1} \exp(-yx^m) \quad (34)$$

$$I_w(x) = 1 - \exp(-yx^m) \quad (35)$$

and the proper expressions for the number and weight averages are

$$\bar{x}_n = \frac{1}{y^{1/m} \Gamma\left(\frac{m+1}{m}\right)} \quad (36)$$

$$\bar{x}_w = \frac{\Gamma\left(\frac{m+1}{m}\right)}{y^{1/m}}, \quad (37)$$

respectively.

(d) The logarithmic-normal (Wesslau) distribution is two-parameter function (parameters x_m and σ), and it is analytically expressed as

$$W(x) = \frac{1}{x\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{\ln x - \ln \bar{x}_n}{\sigma}\right)^2\right] \quad (38)$$

$$I_w(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{\ln x - \ln \bar{x}_n}{\sigma\sqrt{2}}\right)\right] \quad (39)$$

where erf stands for the error function. The corresponding averages are given by the following equations:

$$\bar{x}_n = \bar{x}_m \exp\left(-\frac{\sigma^2}{2}\right) \quad (40)$$

$$\bar{x}_w = \bar{x}_m \exp\left(+\frac{\sigma^2}{2}\right). \quad (41)$$

To choose a proper function for describing the MWD of a polymeric sample is not an easy task. The shape of the molecular weight distribution is determined mainly by the mechanism of polymerization reaction, and partly by the characteristics of the polymer preparation process. Theoretical distribution functions [12–16] are derived on the basis of statistical considerations for the idealized polymerization reactions. Consequently, the experimentally observed deviations from the theory may imply the incorrectness of the theoretical postulates or built-in approximations. The theoretical fundamentals of majority of developed functions are very similar and,

additionally, the mechanism of reaction is not completely known, so the pre-choice of the function is rarely possible. Besides, none of the functions are especially simple or computationally suitable. It is therefore best to take into account several theoretical distribution functions [17–19]. For each of the chosen functions, the independent fitting to the experimental data should be performed numerically, and the optimal parameter values should be computed. To check the validity of the obtained fits, a numerical criterion should be postulated, e.g. the mean-square deviation of the experimental and calculated values. On the basis of the defined criterion, the most adequate theoretical function is chosen. In further analysis, if necessary, the polymer MWD may be replaced by a given function, of course with proper parameter values.

THE POLYDISPERSITY INDEX: PROBLEM, EXAMPLES AND DISCUSSION

The experimental determination of the polymer MWD is routinely performed by the gel-permeation chromatography (GPC) or the size-exclusion chromatography (SEC) method. Nevertheless, the qualitative information on the MWD width may be obtained by direct determination of number and weight MWD averages, usually by the membrane osmometry and light scattering methods, respectively.

A ratio of the molecular weight averages, known as the polydispersity index (heterogeneity ratio, dispersion ratio, nonuniformity coefficient), defined by the expression

$$I = \frac{\bar{x}_w}{\bar{x}_n} = \frac{\bar{M}_w}{\bar{M}_n} \quad (42)$$

is commonly used for description of the polymer MWD. The wide variety of the terms point out a possible misunderstanding of the physical significance of the ratio. The polydispersity index is often taken as the absolute measure of the molecular weight distribution, although some objections have been presented to confront that opinion [20]. The higher polydispersity index indeed implies the wider MWD. However, contrary to the widely accepted belief, the reverse is not true, as will be illustrated below.

In the theory of the probability functions, the variance of distribution σ_d^2 is normally used as the measure of the distribution width. The variance of the MWD may be related to the corresponding MWD averages:

$$\sigma_d^2 = \bar{x}_n^2 \left(\frac{\bar{x}_w}{\bar{x}_n} - 1 \right). \quad (43)$$

Theoretical analysis of the polymer MWD usually involves the comparison of the distributions with the same number average molecular weight [12]. The narrowest possible distribution, which has a certain $\bar{x}_n$, is the monodisperse one. All averages of the monodisperse polymer MWDs are the same, and the polydispersity index is equal to 1. The part of equation (43) in parentheses represents the deviation of the polydispersity index from the value of 1, so every deviation (positive, because $\bar{x}_w$ is always larger than or equal to $\bar{x}_n$) is reflected in the increase of the MWD width. Yet, while transferring the theory into practice,

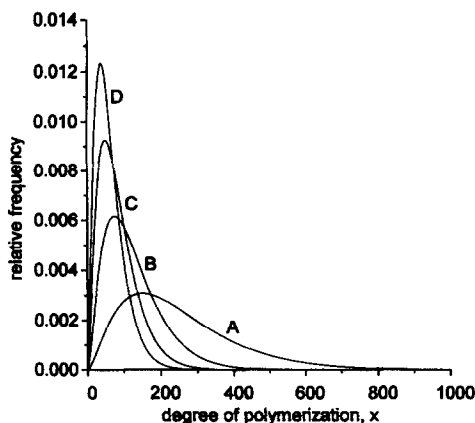


Fig. 1. Simulated Schulz distribution functions with the identical polydispersity index $I = 1.67$. Parameter b for all curves equals 1.5. Parameter α equals 0.01, 0.02, 0.03, 0.04 for curves A, B, C, D, respectively.

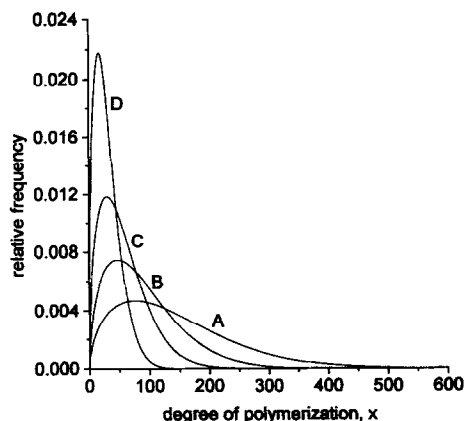


Fig. 3. Simulated Tung distribution functions with the identical polydispersity index $I = 2.42$. Parameter m for all curves equals 1.5. Parameter γ equals $5 \cdot 10^{-4}$, 10^{-3} , $2 \cdot 10^{-3}$, $5 \cdot 10^{-3}$ for curves A, B, C, D, respectively.

the first part of expression (43), i.e. the square of the number average molecular weight, is often ignored. Consequently, the increase in the MWD width is automatically interpreted in terms of the increase in the polydispersity index. Nevertheless, a very simple simulation of the molecular weight distributions can clearly confirm the possible existence of distributions with large differences in width, but with the unique polydispersity index.

Considering the two-parameter distribution functions, the change of any of the parameters undoubtedly causes the change in the distribution width. Nevertheless, the polydispersity index is very often related to the value of only one of the parameters. It can be easily proved for the Schulz's distribution by setting the ratio of the weight to the number average molecular weight [equations (33) and (32)]:

$$I = \frac{b+1}{b} \quad (44)$$

and similarly for the Tung's:

$$I = \Gamma\left(\frac{m+1}{m}\right) \cdot \Gamma\left(\frac{m-1}{m}\right) \quad (45)$$

and logarithmic-normal distribution:

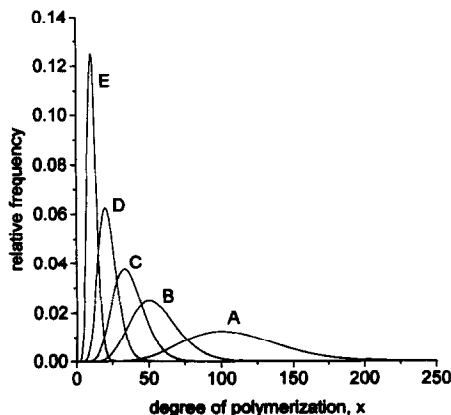


Fig. 2. Simulated Schulz distribution functions with the identical polydispersity index $I = 1.1$. Parameter b for all curves equals 10. Parameter α equals 0.1, 0.2, 0.3, 0.5, 1.0 for curves A, B, C, D, E, respectively.

$$I = \exp(\sigma^2). \quad (46)$$

The examples of the simulated distributions having the same polydispersity index, and various widths are illustrated by Figs 1–4.

Contrary to the common expectations, the large increase of the polydispersity index may affect the distribution width only slightly (Fig. 5). These minor changes are usually of no practical importance. Therefore, for judging the MWD width, it is unacceptable to use exclusively the polydispersity index. Moreover, assigning the larger polydispersity index to the wider distribution may be completely wrong. At least one additional datum (e.g. number or weight average molecular weight, or one of the higher ratios of MWD averages [21–25]) is needed beside the polydispersity index for the correct evaluation of the MWD width. Fortunately, these data are usually either available or easy to determine.

NORMALIZATION OF MOLECULAR WEIGHT DISTRIBUTIONS

On the basis of the MWDs of styrene-acrylonitrile copolymers varying in composition, the existence of distributions possessing the same polydispersity index

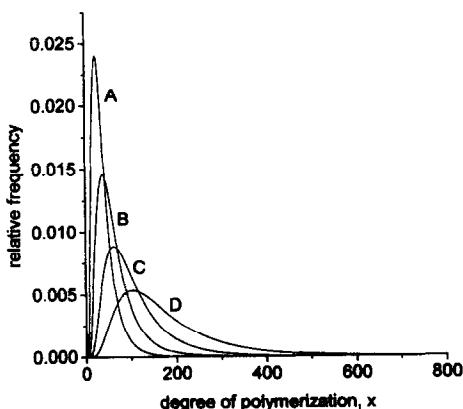


Fig. 4. Simulated logarithmic-normal distribution functions with the identical polydispersity index $I = 1.43$. Parameter σ for all curves equals 0.6. Parameter $\ln \bar{x}_m$ equals 3.5, 4.0, 4.5, 5.0, for curves A, B, C, D, respectively.

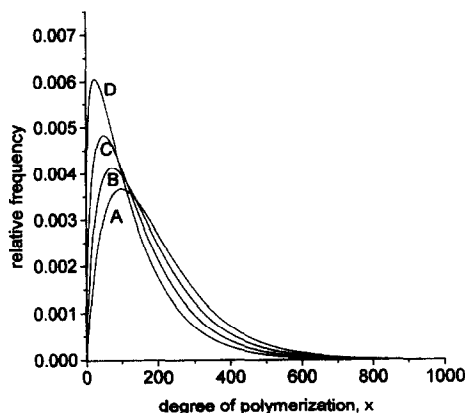


Fig. 5. Simulated Schulz distribution functions with varying polydispersity index $I = 1.67$. Parameter α for all curves equals 0.01. Parameter b equals 0.25, 0.5, 0.7, 1.0, and polydispersity index equals 5.0, 3.0, 2.33, 2.0 for curves A, B, C, D, respectively.

but diverse distribution width is experimentally confirmed. The MWDs of the copolymers are determined by the GPC method. Experimental details are presented in our previous article [19]. Using expressions (20), (21) and (42), the polydispersity indices are calculated. The observed increase in the MWD width with increasing content of acrylonitrile units in the copolymer is not reflected in the change of the polydispersity index, which is rather constant for all investigated copolymers. To quantify the dependence of the distribution width on the composition of copolymers, the MWDs of investigated copolymers are described by Schulz's theoretical distribution function.

Before any comparison of the various MWDs, the processing of the "raw" distribution data is needed. It is worthwhile to present here the entire procedure of normalization, as in practice the normalization is either not performed, it is performed improperly, or it is not described and hence not available to an interested reader.

As already mentioned, the experimentally determined MWDs contain sets of data connecting individual molecular weights (abscissae) with the corresponding relative frequencies (ordinates). The first normalization (normalization no. 1) is performed in order to determine the best-fitting theoretical function, and it comprises the normalization of both the abscissae and the ordinates. The goal of the abscissae normalization is to simplify the fitting procedure, by avoiding large numbers (molecular weights) in numerical operations. Division of all abscissae of all distributions by the same, arbitrarily chosen value brings them into the range of relatively low numbers. The first normalization of ordinates is performed by dividing all ordinates of a particular distribution by the corresponding maximal ordinate. Consequently, the chosen criteria for the success of data fitting (usually mean-square deviations of experimental and calculated data) become comparable values for all experimental distributions. The theoretical distribution function capable of describing all experimental MWDs is chosen on the basis of the criterion values computed during the fitting procedure.

The aim of the following normalization procedures is the comparison of the various experimental distributions via the best-fitting distribution function, chosen previously. The type of the normalization depends on the purpose of the comparison. To compare distribution widths (normalization no. 2), the abscissae of all experimental distributions should be divided by the same, arbitrarily chosen value similarly as for the first described normalization. The computed numerical values of the theoretical distribution function parameters are strictly dependent on the chosen norm, so the norm should be clearly presented. The ordinates of the distributions should be normalized in such a way as to satisfy the basic condition for the distribution, given by equation (2). Namely, the areas under the distribution curves should be equal to 1. It is easy to show that all ordinates of a particular experimental MWD should be divided by the corresponding area, calculated by the equation

$$A = \int_0^{\infty} B(M) dM \quad (47)$$

where A denotes the area and $B(M)$ is the (experimentally determined) relative frequency of the molecules having molecular weight M . The area is calculated by the convenient numerical method, e.g. trapezoidal or Simpson's rule. As a direct consequence, wider distributions appear with lower ordinates, and narrower MWDs have higher ordinates. Afterwards the previously chosen theoretical distribution function may be fitted to the normalized MWD data.

If the objects of the comparison are the distribution shapes (normalization no. 3), disregarding the absolute width, then the abscissae of a particular distribution should be divided by some of the averages, e.g. by the number average molecular weight. In order to avoid the ambiguity of the interpretation, it is again very important to stress the chosen norm. All ordinates of a particular distribution should be divided by the corresponding maximal ordinate. Hence the ordinate heights become comparable, and the eventual variations in shape are more explicit. This normalization procedure may be followed by the convenient fitting procedure as well.

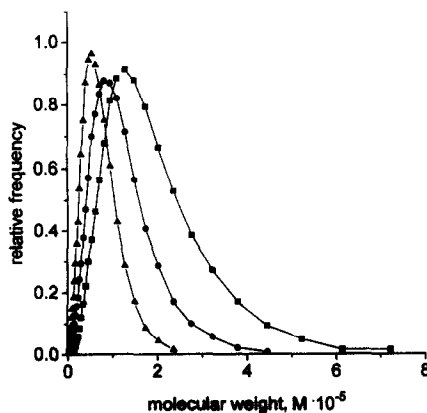


Fig. 6. Non-normalized molecular weight distributions of poly(styrene-co-acrylonitrile)s with varying composition; relative concentration vs molecular weight. Experimental points are connected with solid line (for clarity).

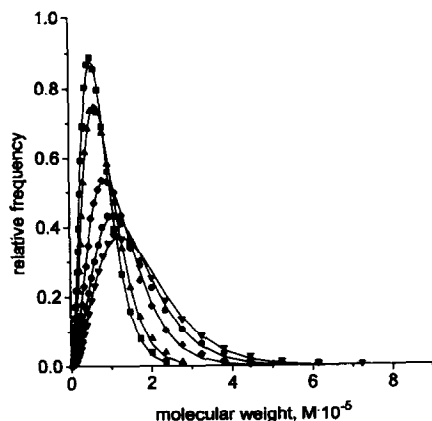


Fig. 7. Molecular weight distributions of poly(styrene-co-acrylonitrile)s after normalization no. 2; relative concentration vs molecular weight. Symbols represent experimental data, solid lines indicate Schulz distribution function fits. Significant change in molecular weight distribution width is not reflected in the polydispersity index change ($I = 1.6$ – 1.7 for all curves).

After selecting Schulz's distribution function as the best-fitting one, the experimental MWDs of styrene-acrylonitrile copolymers (Fig. 6) are normalized in two ways. For the quantitative comparison of distribution widths, normalization no. 2 is applied, and optimal Schulz parameters are determined. The linear relationship between the parameter α and the copolymer composition is observed. The parameter b is approximately constant for all investigated distributions. Obviously the polydispersity index, which is dependent only on the parameter b [equation (44)], will have the constant value, too. Yet the distribution widths significantly change with the composition (Fig. 7). Therefore, the constancy of the polydispersity index for the MWDs varying in width is clearly confirmed. When the normalization procedure no. 3 is performed, using the corresponding number average molecular weights as the norms, the unique "master" distribution curve is obtained (Fig. 8). The shape of the curve is determined by the mechanism of the formation of macromolecules. The

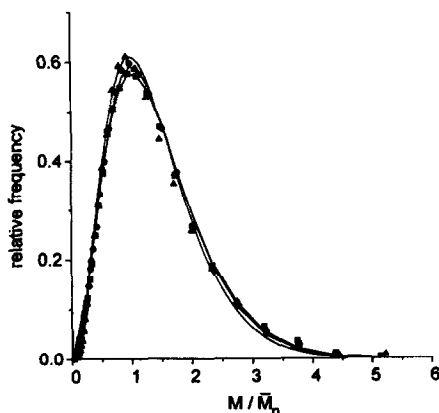


Fig. 8. "Master" distribution curve for the poly(styrene-co-acrylonitrile)s after normalization no. 3; relative concentration vs normalized molecular weight. Symbols represent experimental data, solid lines indicate Schulz distribution function fits.

constancy of the parameter b may be attributed to the unique shape or, consequently, the unique mechanism of copolymerization within the whole experiment. The changes of the copolymer composition are connected with the changes of the MWD width and they are reflected in the variations of the parameter α . As explained elsewhere [19], no chemical composition distribution (CCD) effects [3, 6, 7] are expected to affect the shape of the experimental MWDs, since the CCD is kept reasonably narrow.

The complete MWD, and not only the molecular weight averages, should be determined for the proper characterization of a polymer. The correct MWD data analysis and the approximation of the experimental distribution by some of the well-known theoretical distribution functions, greatly simplifies and accelerates the information transfer and storage. The complete, original distribution may easily be reconstructed from the known parameters of the applied function.

CONCLUSIONS

- (1) The most important characteristics of molecular weight distributions of polymer are briefly presented.
- (2) The importance of the polydispersity index as a measure of the shape and the relative width of the distribution is discussed.
- (3) It is completely unacceptable to draw conclusions on the absolute width of the molecular weight distributions of polymers solely on the basis of polydispersity index, which is contrary to the widely accepted belief.
- (4) The advantages of the theoretical distribution functions for the description of molecular weight distributions of polymers are described.
- (5) For the comparison of molecular weight distributions of polymers, certain data analysis should be necessarily performed. The required procedure is fully presented.

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